

DOUBLY ISOLATED BATYREV MIRROR PAIRS AND NON-SMOOTHABLE CALABI–YAU THREEFOLDS

BERND JOHANNES WUEBBEN

ABSTRACT. We classify reflexive four-polytopes for which the generic anticanonical hypersurfaces associated with the face fans of the polytope and its polar both have at most isolated singularities. Assuming that the database files used contain the complete Kreuzer–Skarke classification without repetition, there are exactly 590 such polytopes. Their two-dimensional faces belong to twelve lattice-isomorphism classes, all triangles, zonotopes, or reflexive polygons. The only nonsmoothable local models are the cyclic quotients $\frac{1}{3}(1, 1, 1)$ and $\frac{1}{5}(1, 1, 3)$ and the anticanonical cone over the first Hirzebruch surface; no singularity has a positive-dimensional reduced deformation space without a smoothing component. Within this classification, the Hirzebruch cone occurs in a unique mirror pair, associated with polytopes having 22 and 26 vertices. We exhibit this pair explicitly: one hypersurface is nonsmoothable, while all singularities of its mirror are locally smoothable. The crepant resolutions have Hodge numbers $(20, 26)$ and $(26, 20)$. At the opposite extreme, the self-dual 24-cell is the unique member of the classification for which both hypersurfaces are smooth before crepant resolution.

1. INTRODUCTION

Let $\Delta \subset N_{\mathbb{R}} = \mathbb{R}^4$ be a reflexive polytope all of whose edges have lattice length one, let $\Delta^\circ \subset M_{\mathbb{R}}$ be its polar dual, and let $X \in |-K_{\mathbb{P}_\Delta}|$ be a generic anticanonical hypersurface in the Gorenstein Fano toric fourfold associated to the face fan of Δ . (Throughout, Δ is the *fan* polytope, Batyrev’s Δ^* , so our Δ° is his Δ ; the same convention is in force in [Wue26a, Wue26b].) By Batyrev [Bat94], X is a Calabi–Yau threefold with Gorenstein canonical singularities. For each two-dimensional face $F \preceq \Delta$ whose cone $C(F)$ is singular, X has exactly $\ell(F^\circ)$ points with analytic germ $C(F)$, and these are all its singularities [Wue26a, Prop. 3.1]. Here $\ell(F^\circ)$ denotes the lattice length of the edge $F^\circ \preceq \Delta^\circ$ dual to F .

The Altmann–Corti–Filip–Petracci trichotomy, recalled as [Wue26a, Thm. 2.2], classifies $C(F)$ by the combinatorics of F into reduced-rigid (R), deformable-but-non-smoothable (D), and smoothable (S). Here *reduced-rigid* means that the reduced miniversal base is a point; first-order deformations may still exist. A single face of the first two types forces X to admit no smoothing, and [Wue26b] studies when the type-(S) configurations smooth.

Mirror symmetry, in the Batyrev picture, is the polar duality $\Delta \leftrightarrow \Delta^\circ$, which for 4-polytopes exchanges two-dimensional faces with edges. This note asks the mirror question: when are X and the mirror $X^\circ \subset \mathbb{P}_{\Delta^\circ}$ *both* isolated-singular (we then call the pair *doubly isolated*), and which local models, in particular which non-smoothable ones, can occur in that situation?

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The entry point is exact and elementary. For a two-dimensional face F write $u_1, u_2 \in M$ for the primitive inner normals of the two facets of Δ containing F , so $F^\circ = \text{conv}\{u_1, u_2\}$, and let $M_F = M \cap (\mathbb{R}u_1 + \mathbb{R}u_2)$. We call the two-dimensional cone $\text{Cone}(u_1, u_2)$ the *normal cone* of F and its dual the *transverse cone* of Δ along F (the local model of Δ in the directions normal to F), and we say Δ is *transversally smooth* along F when this cone is smooth.

Lemma 1.1. $\ell(F^\circ) = [M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2]$, the multiplicity of the normal cone $\text{Cone}(u_1, u_2)$ of F (equivalently, of its dual, the transverse cone of Δ along F). In particular $\ell(F^\circ) = 1$ if and only if the transverse cone is smooth.

The proof is given in Lemma 3.1; Remark 3.2 explains its transverse interpretation.

X is isolated-singular exactly when Δ has unit edges, and X° exactly when every two-dimensional face of Δ has $\ell(F^\circ) = 1$; a face with $\ell(F^\circ) = m \geq 2$ contributes to X° a curve of transverse A_{m-1} singularities (Proposition 4.1(2)). We obtain:

Corollary 1.2. X and X° are both isolated-singular if and only if Δ has unit edges and every two-dimensional face of Δ is transversally smooth. We call such Δ both-sides unit. The condition is self-dual: Δ is both-sides unit if and only if Δ° is.

Our main results determine the both-sides unit polytopes, and the local models they carry, *completely*, subject to one explicit input assumption. We use the following *database hypothesis*: the per-vertex-count files described in §5 contain, without repetition, exactly the members of the Kreuzer–Skarke classification in their stated vertex range; the unique remaining 36-vertex member is included separately. Under this hypothesis the scan covers all 473,800,776 reflexive 4-polytopes [KS00]. Every polytope returned by the scan is then re-verified in exact arithmetic, so the classification statements below have finite computer-assisted proofs conditional only on that database hypothesis. The transverse identity and the assertions about each explicitly displayed polytope are unconditional.

Theorem 1.3 (classification of both-sides local models). *Under the database hypothesis above, there are exactly 590 both-sides unit reflexive 4-polytopes. Every two-dimensional face of every one of them is*

- a triangle (the germ is a cyclic quotient point, or smooth), or
- a zonotope (a centrally-symmetric unit-edge polygon, always of type (S)), or
- a reflexive polygon: one of the five unit-edge polygons with exactly one interior lattice point, i.e. the Fano polygons of the five smooth toric del Pezzo surfaces.

All five reflexive polygons occur. The complete face inventory is given in Table 2.

Theorem 1.4 (the non-smoothable germs of doubly isolated pairs). *Under the database hypothesis, exactly three of the 590 both-sides unit polytopes carry a non-smoothable two-dimensional face. Writing $[k, i]$ for a unit-edge polygon with k edges and i interior lattice points, the germs occurring are:*

- $\frac{1}{3}(1, 1, 1)$ (the cone over the $[3, 1]$ triangle), six times on one 9-vertex polytope;
- $\frac{1}{5}(1, 1, 3)$ (the cone over the $[3, 2]$ triangle), ten times on a single simplex;
- the anticanonical cone over $\mathbb{F}_1$, exactly once, on a unique 22-vertex polytope.

In particular no deformable-but-non-smoothable (D) face occurs on any both-sides unit polytope.

Corollary 1.5. Assume the database hypothesis. Let X be a unit-edge Batyrev Calabi–Yau threefold with a singular point of type (D), for instance the pentagon-cone threefolds of [Wue26a, Thms. 5.3, 5.8]. Then the Batyrev mirror X° is non-isolated: it carries a

curve of transverse A_{m-1} singularities for some $m \geq 2$. The deformable-but-non-smoothable phenomenon never occurs on a doubly isolated mirror pair.

For the remaining non-quotient case, write dP_d for the smooth toric del Pezzo surface of degree d , with $d = 6, 7$.

Theorem 1.6 (the unique $\mathbb{F}_1$ mirror pair). *Under the database hypothesis, up to lattice isomorphism there is exactly one doubly isolated Batyrev mirror pair whose singularities are not all quotient points or smoothable: the pair $(\Delta_{\mathbb{F}_1}, \Delta_{\mathbb{F}_1}^\circ)$ of Theorem 6.1, with 22 and 26 vertices. The generic anticanonical $X \subset \mathbb{P}_{\Delta_{\mathbb{F}_1}}$ has 20 ordinary double points, one dP_6 -cone point, and one point whose germ is the anticanonical cone over $\mathbb{F}_1$; since that germ admits no smoothing at all, X admits no smoothing [Wue26a, Cor. 4.3]. Its mirror X° has 26 ordinary double points, two dP_7 -cone points and two dP_6 -cone points; every singular germ of X° is of type (S). The maximal projective crepant partial resolutions have $(h^{1,1}, h^{2,1}) = (20, 26)$ and $(26, 20)$ respectively.*

Thus a nonsmoothable Calabi–Yau threefold can have an isolated-singular Batyrev mirror with every singular point locally smoothable. Every dual edge on the mirror side has length one, so the single-face construction of [Wue26b] does not apply. The 26 diagonal-relation vectors of its nodes have rank 18 and two coloops: the coefficients of those two vectors vanish in every relation. Hence there is no relation with every coefficient nonzero. This nodal calculation alone does not decide smoothability in the presence of the four del Pezzo-cone points. The mixed necessity theorem of [Wue26d] proves that the mirror is also nonsmoothable, as discussed in Remark 6.3.

Theorems 1.3, 1.4 and 1.6 answer the mirror question of [Wue26a, §6.4] for the non-smoothable local models: among them, only the cyclic quotients and the $\mathbb{F}_1$ -cone, including the germs of Gross’s two exceptional surfaces $\mathbb{P}^2$ and $\mathbb{F}_1$ [Gro97, Thm. 5.8], admit doubly isolated mirror realizations, and the $\mathbb{F}_1$ -cone does so exactly once in the entire toric hypersurface landscape.

Under the same database hypothesis, the opposite extreme of the classification carries a matching uniqueness. Exactly eight of the 590 polytopes have all two-dimensional faces smooth, so that X itself is smooth; for exactly one of them the mirror is smooth as well: the self-dual 24-cell, whose hypersurface is the classical self-mirror threefold with $(h^{1,1}, h^{2,1}) = (20, 20)$ [Bra12]. The smooth Batyrev mirror pair is thus unique in the entire classification (Example 5.4).

The first asymmetric faces. Call a polygon *asymmetric* when it is neither a triangle nor centrally symmetric. Under the database hypothesis, no such face occurs on a both-sides unit polytope with at most 18 vertices, a range containing 395,406,329 reflexive 4-polytopes (83.5% of the classification). The first asymmetric face occurs at 19 vertices. The complete inventory consists of the dP_7 pentagon (87 faces, on polytopes with 19 to 31 vertices) and the $\mathbb{F}_1$ quadrilateral (once). Section 5.3 gives the first example and explains why a bound on the number of vertices does not respect polar duality.

All mathematical predicates and reported invariants in the scan, the exact re-verification of all 590 polytopes, and the mirror-pair calculation use integer or rational arithmetic, including the cyclic ordering of planar faces. The programs build on those of [Wue26a] and are included as ancillary files; `both_sides_census.py` reconstructs the finite face inventories, counts, and Hodge numbers from the recorded polytopes. The code and the scan data are public at <https://github.com/bwuebben/calabi-yau-smoothability>, where every count in this note is recomputable from `src/both_sides_census.py`.

Acknowledgements. This note continues the series [Wue26a, Wue26b]; the question of smoothing canonical Calabi–Yau singularities was posed to the author by Mark Gross many years ago.

2. PRELIMINARIES

We keep the notation of [Wue26a]. For a lattice polygon Q with unit edges placed at height one, $C(Q)$ denotes the associated Gorenstein toric affine threefold. A unimodular triangle gives a smooth cone and is excluded from the following trichotomy. For every other unit-edge polygon, Q is of type (R), (D) or (S) according as its edge-vector multiset has no proper nonempty zero-sum sub-multiset, has one but no partition into opposite primitive pairs and unimodular zero-sum triples, or has such a partition [Alt97, CFP22]. Here a zero-sum triple (a, b, c) is unimodular when $|\det(a, b)| = 1$. Type (R) cones are *reduced-rigid*: their reduced miniversal base is a point, although first-order deformations may exist. Type (S) polygons are exactly those with singular smoothable cone; a single germ of type (R) or (D), one whose miniversal base has no smoothing component, obstructs every global smoothing of X [Wue26a, Lemma 4.1, Cor. 4.3]. The notation $[k, i]$ records the number k of edges and the number i of interior lattice points of a unit-edge polygon; these two invariants do not determine its lattice-isomorphism class. The kind column of Table 2 distinguishes the two classes with invariants $[4, 1]$.

A unit-edge polygon is a *zonotope* (a Minkowski sum of unit segments) if and only if it is centrally symmetric, and every zonotope is of type (S). A unit-edge polygon with exactly one interior lattice point is reflexive. Indeed, translate that point to the origin and let the polygon have k edges. Pick’s theorem gives normalized area k , while the fan from the origin decomposes it into k lattice triangles of positive integral normalized areas, one over each unit edge. Every one therefore has normalized area one, so the origin has lattice distance one from every edge. There are five, the Fano polygons of $\mathbb{P}^2$, $\mathbb{F}_1$, $\mathbb{P}^1 \times \mathbb{P}^1$, dP_7 , dP_6 (the subscript is the degree: $\text{dP}_7 = \text{Bl}_2 \mathbb{P}^2$, $\text{dP}_6 = \text{Bl}_3 \mathbb{P}^2$), the cone $C(Q)$ being the anticanonical cone over the corresponding surface. Of the five, the $\mathbb{P}^2$ triangle and the $\mathbb{F}_1$ quadrilateral are reduced-rigid and the other three are of type (S) [Wue26a, Rem. 2.6]. For a reflexive 4-polytope Δ we write $\Delta^\circ = \{u : \langle u, v \rangle \geq -1 \ \forall v \in \Delta\}$; a face $G \preceq \Delta$ has dual face $G^\circ \preceq \Delta^\circ$ with $\dim G + \dim G^\circ = 3$. By [Wue26a, Prop. 3.1], for Δ with unit edges the generic $X \in |-K|$ has, over each two-dimensional face F with $C(F)$ singular, exactly $\ell(F^\circ)$ singular points with germ $C(F)$, and no other singular points; and an edge of Δ of lattice length $m \geq 2$ produces a curve of transverse A_{m-1} singularities [Wue26a, Rem. 3.2]. Throughout, $\Delta \subset N_{\mathbb{R}}$ is the *fan* polytope, Batyrev’s Δ^* , so our Δ° is his Δ ; the same convention is used in [Wue26a, Wue26b].

3. THE TRANSVERSE-MULTIPLICITY IDENTITY

Let F be a codimension-two face of a reflexive polytope $\Delta \subset N_{\mathbb{R}}$ of arbitrary dimension, and let $u_1, u_2 \in M$ be the primitive inner normals of the two facets of Δ containing F , so that $\langle u_i, \cdot \rangle = -1$ on F and $F^\circ = \text{conv}\{u_1, u_2\}$.¹ Put $M_F = M \cap (\mathbb{R}u_1 + \mathbb{R}u_2)$, a rank-two saturated sublattice of M containing u_1, u_2 as primitive vectors.

Lemma 3.1. $\ell(F^\circ) = [M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2]$.

Proof. Both sides depend only on the pair (u_1, u_2) inside the rank-two lattice M_F . Choose a $\mathbb{Z}$ -basis of M_F in which $u_1 = (0, -1)$; this is possible because u_1 is primitive. Write

¹The u_i are vertices of Δ° ; [Wue26a] and the ancillary programs list the sign-flipped functionals taking the value +1 on the facet. Lattice lengths are unaffected.

$u_2 = (p, q)$; here $p \neq 0$, since $p = 0$ would force $u_2 = \pm u_1$, impossible as $F^\circ = \text{conv}\{u_1, u_2\}$ is an edge of Δ° and 0 is interior to Δ° . Then $p = \det(u_1, u_2) = \pm[M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2]$, and $\ell(F^\circ) = \ell(\text{conv}\{u_1, u_2\}) = \gcd(u_2 - u_1) = \gcd(p, q + 1)$, the two gcds agreeing because M_F is saturated in M , so lattice length in M may be computed in the chosen basis of M_F .

Let v be a vertex of F and let $\bar{v} \in N_F := N/(N \cap u_1^\perp \cap u_2^\perp)$ be its image; since u_1, u_2 vanish on $N \cap u_1^\perp \cap u_2^\perp$ they descend to elements of $M_F = N_F^\vee$, and $\bar{v}$ is a lattice point of $N_F \cong \mathbb{Z}^2$ with $\langle u_i, \bar{v} \rangle = -1$. Writing $\bar{v} = (a, b)$ in the basis of N_F dual to the chosen basis of M_F , $\langle u_1, \bar{v} \rangle = -b = -1$ gives $b = 1$, and $\langle u_2, \bar{v} \rangle = pa + q = -1$ gives $pa = -1 - q$, so $p \mid (q + 1)$. Hence $\gcd(p, q + 1) = |p|$, and $\ell(F^\circ) = |p| = [M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2]$. $\square$

Remark 3.2. Lemma 3.1 is elementary and may be known; what matters here is the reading it provides. The proof works in every dimension for a codimension-two face. The right-hand side is the multiplicity of the two-dimensional cone $\text{Cone}(u_1, u_2)$, which is the normal cone of F and dually the transverse cone of Δ along F (for two-dimensional cones the multiplicities of a cone and its dual agree). Thus *the dual-edge length counts exactly the transverse singularity multiplicity*, and $\ell(F^\circ) = 1$ is the statement that Δ is transversally smooth along F . (Transversal smoothness concerns only the directions normal to F : the germ $C(F)$ itself may well be singular.) The lattice-point hypothesis on v is essential: for an arbitrary pair of primitive vectors the two sides can differ.

Corollary 1.2 is immediate: X is isolated-singular iff Δ has unit edges; X° is isolated-singular iff Δ° has unit edges, i.e. every edge F° of Δ° has $\ell(F^\circ) = 1$, i.e. by Lemma 3.1 every two-dimensional face of Δ is transversally smooth. Self-duality follows since the conditions “unit edges” and “all dual edges of length one” are exchanged by $\Delta \leftrightarrow \Delta^\circ$.

4. THE MIRROR CORRESPONDENCE OF THE SINGULAR STRATA

Proposition 4.1. *Let Δ be reflexive with unit edges and F any two-dimensional face. Under polar duality F corresponds to the edge F° of Δ° , and:*

- (1) *if $C(F)$ is singular, X has $\ell(F^\circ)$ isolated points of type $C(F)$ over F (and none over F otherwise);*
- (2) *X° has, along the surface $V(\sigma_{F^\circ}) \subset \mathbb{P}_{\Delta^\circ}$ (the orbit closure of the cone σ_{F° over the edge F° in the face fan of Δ°), a curve of transverse $A_{\ell(F^\circ)-1}$ singularities (and none if $\ell(F^\circ) = 1$).*

In particular X° is isolated-singular if and only if $\max_F \ell(F^\circ) = 1$, i.e. Δ is both-sides unit.

Proof. (1) is [Wue26a, Prop. 3.1]. For (2), the edge F° of Δ° has lattice length $\ell(F^\circ)$; an edge of lattice length m produces on the generic anticanonical hypersurface a curve of transverse A_{m-1} singularities (§2), and none when $m = 1$. Hence X° is isolated-singular iff every two-dimensional face of Δ has $\ell(F^\circ) = 1$. $\square$

Thus the $\ell(F^\circ)$ singular points of X over F and the $A_{\ell(F^\circ)-1}$ -curve of X° correspond under polar duality, and the two threefolds are simultaneously isolated-singular exactly on the both-sides unit polytopes. Whenever Δ has *any* face with $\ell(F^\circ) \geq 2$ (the generic situation), the mirror is non-isolated and its singularities leave the isolated framework of [Alt97, CFP22]. Filip [Fil26] constructs, from a Laurent polynomial, a distinguished formal deformation of the associated non-isolated Gorenstein toric pair and proves that its general fibre is smooth exactly when the polynomial is 0-mutable; this does not classify all components of every non-isolated germ.

5. ENUMERATION OVER THE KREUZER–SKARKE CLASSIFICATION

5.1. Methodology. The both-sides unit condition is a condition on faces: all edges of lattice length one and, by Lemma 3.1, all dual edges of length one. We tested it over the per-vertex-count copy of the Kreuzer–Skarke classification used in [Wue26a, §6.1], with the production implementation built for that scan. Subject to the database hypothesis in the introduction, this covers the complete classification. All facet incidences, lattice lengths, determinants, and final predicates are checked in exact integer arithmetic, with no heuristic pruning. Planar vertices are ordered by an exact half-plane and cross-product comparator, so this step also uses no floating-point arithmetic. For reproducibility, the ancillary manifest pins the database repository at revision `60c0e119a036` (with the full identifier in the manifest) and gives the byte size and SHA-256 digest of every input file. The resumable driver verifies those digests and validates the row count, result schema, and every recorded positive hit before accepting an existing result; new results are written atomically, and every invocation records a transcript. These controls fix the identity and integrity of the finite input and saved artifacts, but do not remove the database hypothesis or independently recheck every negative row. Implementation correctness is controlled in three complementary ways. (i) Before scanning each file of the database, the implementation was re-checked against an independently written reference implementation on random samples from that file. (ii) The 3,905,789 polytopes with at most nine vertices were scanned in full by both implementations, with identical results. (iii) The 590 polytopes found pass a global consistency check under polar duality: the both-sides condition is self-dual (Corollary 1.2), and the polar of every polytope found, computed exactly, is again both-sides unit and agrees with one of the 590 in vertex count, facet count and complete face inventory, an invariance that would generically fail for a false negative whose polar partner was detected. An invariant collision (another member of the 590 sharing its vertex count, facet count and face inventory) could in principle mask a miss, so this control is supporting evidence beside (i) and (ii), not an independent proof. The one polytope missing from the copy of the database we used, which is organized by vertex count, is the 36-vertex hexagon product (see [Wue26a, §6.1]); it is included. Every polytope passing the scan is then re-verified, and its faces classified, by the reference implementation; the ancillary scripts reproduce the full scan from the public database. The scan results are included as ancillary data files, and the program `both_sides_census.py` re-derives every statement of Theorems 1.3, 1.4 and 1.6 from them.

5.2. The classification. Tables 1 and 2 give the vertex distribution and face inventory under the database hypothesis.

vertices	5	6	8	9	10	12	15	16	17	18	19	20	21
#	2	1	1	1	1	1	1	4	4	7	12	24	37
vertices	22	23	24	25	26	27	28	29	30	31	33	36	
#	69	84	98	85	64	42	24	13	9	4	1	1	

TABLE 1. The 590 both-sides unit reflexive 4-polytopes by vertex count (no others occur; in particular none with 7, 11, 13, 14, 32, 34 or 35 vertices). The population is closed under polar duality (Corollary 1.2) and peaks at 24 vertices.

face $[k, i]$	kind	count	germ of X over it
$[3, 0]$	triangle	25,873	smooth point
$[3, 1]$	triangle	6	$\frac{1}{3}(1, 1, 1)$ (R)
$[3, 2]$	triangle	10	$\frac{1}{5}(1, 1, 3)$ (R)
$[4, 0]$	zonotope	15,652	ordinary double point
$[4, 1]$	zonotope	99	cone over $\mathbb{P}^1 \times \mathbb{P}^1$
$[4, 1]$	<i>asymmetric</i>	1	cone over $\mathbb{F}_1$ (R)
$[4, 2]$	zonotope	9	type (S)
$[5, 1]$	<i>asymmetric</i>	87	cone over dP_7
$[6, 1]$	zonotope	395	cone over dP_6 , 2 smoothing components
$[6, 2]$	zonotope	2	type (S)
$[6, 7]$	zonotope	3	type (S)
$[8, 4]$	zonotope	1	type (S), 3 smoothing components

TABLE 2. Every two-dimensional face of every both-sides unit polytope, by class, under the stated database hypothesis (k = edges, i = interior points). Type (R) means that the reduced miniversal base is a point, while type (S) means that the singular germ admits a smoothing. “Asymmetric” = neither a triangle nor centrally symmetric; only the two non-centrally-symmetric reflexive polygons occur, and each germ appears on X exactly once per face since all dual edges have length one. Type-(S) germs have one smoothing component except where noted.

Proof of Theorems 1.3 and 1.4. Both theorems are exhaustive verifications, summarized in Tables 1 and 2; the completeness of the underlying enumeration is the content of §5.1. Granting it, the scan of all 473,800,776 polytopes yielded 590 both-sides unit ones; exact re-analysis of each classifies all their two-dimensional faces into the twelve classes of Table 2, each of which is a triangle, a zonotope, or one of the five reflexive polygons of the smooth toric del Pezzo surfaces. Explicit $GL_2(\mathbb{Z})$ equivalences identify the $[4, 1]$ zonotope with the $\mathbb{P}^1 \times \mathbb{P}^1$ diamond, the $[6, 1]$ zonotope with the dP_6 hexagon, and the $[3, 1]$ triangle with the $\mathbb{P}^2$ triangle. The asymmetric $[5, 1]$ and $[4, 1]$ classes are the dP_7 pentagon and the $\mathbb{F}_1$ quadrilateral. The non-smoothable classes occurring are the rigid triangles $[3, 1]$, $[3, 2]$ and the $\mathbb{F}_1$ quadrilateral; the $[3, 2]$ faces are the ten faces of the quotient simplex of Example 5.2, the $[3, 1]$ faces lie on the 9-vertex polytope of Example 5.2, and the $\mathbb{F}_1$ face on the unique polytope of Theorem 6.1. No face of type (D) occurs. $\square$

Proof of Corollary 1.5. A type-(D) point of X is the cone over a type-(D) face F . By Theorem 1.4, Δ is then not both-sides unit, so some face F' has $\ell(F'^\circ) \geq 2$ and Proposition 4.1(2) gives a singular curve on X° . $\square$

5.3. The first asymmetric examples. Under the database hypothesis, Table 2 shows that 88 asymmetric faces occur, on 55 polytopes. The smallest example has 19 vertices:

$$\begin{aligned} \Delta_{19} = \text{conv}\{ & (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, -1, 0), (0, -1, 0, 0), \\ & (1, -1, -1, 0), (0, 0, 0, 1), (-1, 1, 1, -1), (0, -1, 0, 1), (0, 0, -1, 1), \\ & (-1, 1, 0, -1), (-1, 0, 1, -1), (-1, 0, 1, 0), (-1, 1, 0, 0), (0, -1, -1, 0), \\ & (0, -1, -1, 1), (-1, 0, 0, -1), (-2, 1, 1, -1), (-1, 0, 0, 1)\}, \end{aligned}$$

a both-sides unit polytope one of whose faces is the dP_7 pentagon; its hypersurface is a Calabi–Yau threefold with 14 nodes and one dP_7 -cone point, all with isolated-singular mirror. That face is

$$P = \operatorname{conv}\{(0, -1, -1, 0), (0, -1, -1, 1), (-1, 0, 0, -1), (-2, 1, 1, -1), (-1, 0, 0, 1)\},$$

the fifteenth to nineteenth entries of the list, lying in the two facets with inner normals $u_1 = (1, 0, 1, 0)$ and $u_2 = (1, 1, 0, 0)$; their difference $(0, -1, 1, 0)$ is primitive, so $\ell(P^\circ) = 1$. In the induced lattice its edge multiset is $\{(0, -1), (1, -1), (1, 0), (-1, 2), (-1, 0)\}$, carried by $M = \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix} \in GL_2(\mathbb{Z})$ onto $\{(0, -1), (1, -1), (1, 1), (-1, 1), (-1, 0)\}$, the dP_7 pentagon of [Wue26a, Rem. 2.6]: the unique unit-edge class with five vertices and one interior lattice point.

An earlier conjecture excluded asymmetric faces by a proposed local condition: an asymmetric face with dual-edge length one, all faces of whose star have unit edges, would force another face in its star to have dual-edge length at least two. The condition holds in the 38,571 candidate configurations among the 1,037,834 polytopes with at most eight vertices, but Δ_{19} disproves it. Theorem 1.3 gives the complete face inventory, whose geometric explanation is the subject of Question 7.1.

Remark 5.1 (why not “simplicial”). The stronger guess that a both-sides unit polytope is simplicial fails for a structural reason: the condition is self-dual, and “simplicial and simple” forces a simplex, contradicting the existence of both-sides unit polytopes with more than five vertices. The non-triangular faces are genuine; the content of Theorem 1.3 is that they are zonotopes or reflexive.

Example 5.2 (the low-vertex stratum). Among the 3,905,789 polytopes with at most nine vertices exactly five are both-sides unit; four of them form two polar pairs. One pair is the simplex with smooth anticanonical hypersurface and the simplex

$$\operatorname{conv}\{(1, 0, 0, 0), (0, 1, 0, 0), (2, 4, 5, 0), (3, 3, 0, 5), (-6, -8, -5, -5)\},$$

whose ten faces are the $[3, 2]$ rigid triangles of Table 2 (quotient points of type $\frac{1}{5}(1, 1, 3)$) and whose polar gives a smooth hypersurface. The ambient face-fan variety of that polar is singular at its torus-fixed points, which the generic hypersurface avoids. A second pair consists of a simplicial six-vertex polytope with all faces smooth and its nine-vertex polar, carrying the six $\frac{1}{3}(1, 1, 1)$ triangles and nine $[4, 2]$ zonotopes. In each pair, the non-smoothable member mirrors a *smooth* threefold. The fifth polytope has eight vertices and sixteen facets, and all thirty-two of its faces are smooth triangles, so its hypersurface X is smooth. It shows that closedness under polar duality (Corollary 1.2) does not respect bounds on the number of vertices: its polar is a sixteen-vertex both-sides unit polytope all twenty-four of whose faces are $[4, 1]$ diamonds, so the mirror X° carries twenty-four cone-over- $\mathbb{P}^1 \times \mathbb{P}^1$ points.

Example 5.3 (the top of the classification). The 36-vertex product $H \times H$ of two reflexive hexagons is the reflexive 4-polytope with the most vertices and the one polytope absent from the copy of the Kreuzer–Skarke database used in [Wue26a]. It is both-sides unit; its faces are 36 unit squares and 12 dP_6 hexagons, so its hypersurface has 48 singular points, all of type (S), while its polar, the 12-vertex free sum $H^\circ \oplus H^\circ$, has 72 smooth triangular faces and smooth hypersurface.

Example 5.4 (the smooth mirror pair). Under the database hypothesis, exactly eight of the 590 polytopes have every two-dimensional face smooth, so that X itself is smooth. For exactly one of them is the polar of the same kind: the 24-cell, self-dual, with 24 octahedral facets,

96 unimodular triangular faces, and no lattice points besides the origin and its 24 vertices. Its generic anticanonical hypersurface is the classical self-mirror Calabi–Yau threefold with $(h^{1,1}, h^{2,1}) = (20, 20)$ and $\chi = 0$; the $(1, 1)$ -threefolds of Braun [Bra12] are free quotients of special symmetric members of this family. Under the same hypothesis, the scan shows uniqueness: among all 473,800,776 reflexive 4-polytopes, the 24-cell is the only one whose generic anticanonical hypersurface and mirror are *both* smooth.

6. THE UNIQUE $\mathbb{F}_1$ MIRROR PAIR

Theorem 6.1. *Let*

$$\begin{aligned} \Delta_{\mathbb{F}_1} = \text{conv}\{ & (1, 0, 0, 0), (0, 1, 0, 0), (1, -1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1), (0, 0, 1, -1), \\ & (0, 0, -1, 1), (0, 0, 0, -1), (0, 0, -1, 0), (-1, 1, 0, 0), (0, -1, 0, 0), (-1, 1, -1, 1), \\ & (0, -1, 0, -1), (0, -1, -1, 0), (-1, 1, -1, 0), (-1, 0, 0, -1), (-1, 0, -1, 1), \\ & (-1, -1, 0, -1), (-2, 1, -1, 1), (-2, 1, -1, 0), (-1, -1, -1, 0), (-2, 0, -1, 0)\} \subset \mathbb{Z}^4. \end{aligned}$$

Then $\Delta_{\mathbb{F}_1}$ is a both-sides unit reflexive polytope with 22 vertices and 26 facets, and:

- (1) *its 72 two-dimensional faces are 50 smooth triangles, 20 unit squares, one dP_6 hexagon, and one quadrilateral $GL_2(\mathbb{Z})$ -equivalent to the $\mathbb{F}_1$ polygon; every dual edge has lattice length one;*
- (2) *the generic anticanonical $X \subset \mathbb{P}_{\Delta_{\mathbb{F}_1}}$ is a Calabi–Yau threefold whose singular locus consists of 20 ordinary double points, one dP_6 -cone point, and one point with germ the anticanonical cone over $\mathbb{F}_1$; since one germ without a smoothing component obstructs every global smoothing (§2), X admits no smoothing [Wue26a, Cor. 4.3];*
- (3) *the polar $\Delta_{\mathbb{F}_1}^\circ$ (26 vertices, 22 facets) is both-sides unit with 68 two-dimensional faces: 38 smooth triangles, 26 unit squares, two dP_7 pentagons and two dP_6 hexagons; the mirror X° has 26 ordinary double points, two dP_7 -cone points and two dP_6 -cone points, each with a local smoothing;*
- (4) *the maximal projective crepant partial resolutions satisfy $(h^{1,1}, h^{2,1})(\widehat{X}) = (20, 26)$ and $(h^{1,1}, h^{2,1})(\widehat{X}^\circ) = (26, 20)$;*
- (5) *under the database hypothesis, $\Delta_{\mathbb{F}_1}$ is the only both-sides unit reflexive 4-polytope, up to lattice isomorphism, carrying a non-smoothable face that is not a triangle.*

Proof. (1)–(4) are finite checks on the explicitly listed polytope, a face enumeration reproducible in any lattice-polytope package, carried out in exact arithmetic by the ancillary files.² We record the $\mathbb{F}_1$ face itself, since it is the object the statement is about. Its four vertices are the fourth, eighteenth, nineteenth and twenty-second entries of the list above,

$$F = \text{conv}\{(0, 0, 1, 0), (-1, -1, 0, -1), (-2, 1, -1, 1), (-2, 0, -1, 0)\},$$

and the two facets of $\Delta_{\mathbb{F}_1}$ containing F have inner normals

$$u_1 = (1, 1, -1, -1), \quad u_2 = (1, 0, -1, 0),$$

in the normalization of §1: both take the value -1 at each of the four vertices of F and are ≥ -1 on $\Delta_{\mathbb{F}_1}$. Their difference $u_1 - u_2 = (0, 1, 0, -1)$ is primitive, so $\ell(F^\circ) = 1$. In the induced lattice on the affine span of F the edge multiset is $\{(0, -1), (1, -1), (1, 1), (-2, 1)\}$, and

$$M = \begin{pmatrix} -1 & -1 \\ -1 & 0 \end{pmatrix} \in GL_2(\mathbb{Z}), \quad \det M = -1,$$

²Ancillary files `both_sides_census.py` and `face_data.py`.

carries it onto $\{(-2, -1), (0, -1), (1, 0), (1, 2)\}$, the edge multiset of the $\mathbb{F}_1$ polygon of [Wue26a, Rem. 2.6]; so $C(F)$ is the anticanonical cone over $\mathbb{F}_1$.

The dP_6 hexagon of (1) is recorded the same way. It is the face

$$G = \mathrm{conv}\{(0, 0, -1, 0), (0, -1, -1, 0), (-1, 1, -1, 0), \\ (-2, 1, -1, 0), (-1, -1, -1, 0), (-2, 0, -1, 0)\}$$

(the ninth, fourteenth, fifteenth, twentieth, twenty-first and twenty-second entries of the list), lying in the two facets with inner normals $u_1 = (0, 0, 1, 0)$ and $u_2 = (0, 0, 1, 1)$, whose difference $(0, 0, 0, -1)$ is primitive, so $\ell(G^\circ) = 1$. Its edge multiset in the induced lattice is already the standard hexagon,

$$E(G) = \{(0, -1), (1, -1), (1, 0), (0, 1), (-1, 1), (-1, 0)\},$$

so no change of basis is needed: $C(G)$ is the anticanonical cone over dP_6 , the class with two smoothing components in the catalogue of [Wue26a, Rem. 2.6]. The Hodge numbers are given by Batyrev’s formulas [Bat94, §4], evaluated in exact arithmetic. The mirror swap in (4) is forced by Batyrev duality and serves as a consistency check. Conditional on the database hypothesis, (5) is Theorem 1.4: the scan finds no other non-triangle non-smoothable face on any both-sides unit polytope. $\square$

Remark 6.2 (a threefold with no smoothing whose mirror has only locally smoothable singularities). Theorem 6.1 pairs a Calabi–Yau threefold that cannot be smoothed with a mirror all of whose singularities are locally smoothable. All $26 + 2 + 2$ singular points of the mirror lie over faces with $\ell(F^\circ) = 1$, beyond the single-face construction of [Wue26b]. The node subsystem has no relation with every coefficient nonzero (Remark 6.3).

Remark 6.3 (the node subsystem). Let $q_1, \dots, q_{26}$ be the vertices of $\Delta_{\mathbb{F}_1}^\circ$. For each of its 26 unit-square faces $S = (q_a, q_b, q_c, q_d)$ in cyclic order, form the diagonal-relation vector

$$\rho_S = e_a - e_b + e_c - e_d \in \mathbb{Q}^{\mathrm{Vert}(\Delta_{\mathbb{F}_1}^\circ)}.$$

Exact row reduction of the resulting 26×26 matrix gives rank 18. Exactly two rows are coloops. The corresponding square faces, displayed in cyclic order, are

$$S_- = \mathrm{conv}\{(-1, -1, 0, -1), (-1, 0, 0, -1), (0, 1, 0, -1), (0, 0, 0, -1)\}, \\ S_+ = \mathrm{conv}\{(-1, -1, 0, 1), (0, -1, 0, 1), (0, 0, 0, 1), (-1, 0, 0, 1)\}.$$

Deleting either row lowers the rank to 17, deleting both lowers it to 16, and deleting any other row leaves rank 18. Thus every linear dependence among the 26 node rows has coefficient zero on both ρ_{S_-} and ρ_{S_+} ; in particular there is no dependence with all coefficients non-zero. The calculation is exact and is reproduced by the ancillary row-reduction script. ³

For a hypothetical all-nodal threefold this fails the Friedman–Batyrev–Kreuzer condition recalled in [Wue26b, §7]. It does *not* decide the actual X° : its four additional del Pezzo-cone points have divisorial crepant resolutions, and their local smoothing directions may contribute to the mixed local-to-global obstruction map. The mixed necessity theorem of [Wue26d, Theorem 6.1] requires a relation among the classes in $H_2(L_p; \mathbb{Q})$ of the singularity links whose image in $H_2(\widehat{X}^\circ; \mathbb{Q})$ is zero, whose cone-point components belong to the kernels $\ker(H_2(L_p; \mathbb{Q}) \rightarrow H_2(M_p; \mathbb{Q}))$ for the chosen local Milnor fibres, and whose node coefficients are all nonzero. The resolution $\widehat{X}^\circ$ uses small resolutions at the nodes and blows up the cone points. In this example the two coloop coefficients vanish on every such relation, for

³paper3.node_relations.py.

either smoothing component at each degree-six point [Wue26d, §5.3, Theorem 6.5]. Hence X° is nonsmoothable. Both members of the pair are therefore nonsmoothable, although every singularity of X° is locally smoothable.

7. QUESTIONS

Question 7.1. Can one characterize the unit-edge lattice polygons that occur as two-dimensional faces of reflexive polytopes whose edges and polar edges all have lattice length one? In dimension four, Theorem 1.3 gives the twelve classes in Table 2, subject to the database hypothesis. A geometric explanation would account for the occurrence of only triangles, zonotopes, and reflexive polygons without using the classification. The local condition discussed in §5.3 is insufficient.

Question 7.2. Corti–Filip–Petracci conjecturally relate smoothing components of $C(Q)$ to 0-mutable Laurent polynomials with Newton polygon Q [CFP22], and Filip [Fil26] constructs distinguished formal deformations in the non-isolated case, detecting when their general fibres are smooth. In that language, what distinguishes the twelve face classes of Table 2 — and what is the mirror-theoretic meaning of the unique $\mathbb{F}_1$ face? One would like to read Theorem 1.4 off the canonical scattering diagram of the mirror.

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BERND JOHANNES WUEBBEN, NEW YORK, NY, wuebben@gmail.com